

Snake Sightings Workflow

User Guide

The Taylor Ashe Antivenom Foundation (TAAF)

Powered by Ecoscope · Wildlife Dynamics

This guide walks you through configuring and running the Snake Sightings Workflow, which processes iNaturalist snake observation data from EarthRanger and produces distribution maps, a species summary table, and a structured Word report.

Overview

The Snake Sightings Workflow connects to EarthRanger, fetches iNaturalist snake observation events within a user-defined time range, and produces a complete distribution analysis with maps, a summary table, and a printable Word report.

The workflow produces:

- An **All-Species Distribution Map** — every verified observation with a convex-hull range overlay.
- An **Observation Effort Map** — 15 km grid coloured by total observation count.
- A **Species Richness Map** — 15 km grid coloured by number of unique species per cell.
- A **Species Summary Table** — counts, observers, locations, date ranges, and range area per species.
- Per-species distribution maps — one interactive map and PNG per species, generated dynamically at runtime.
- A **Word Report** (taaf_snake_distribution_report.docx) with all maps, the summary table, and per-species pages.
- An **Interactive Dashboard** with all map widgets and the sortable summary table.

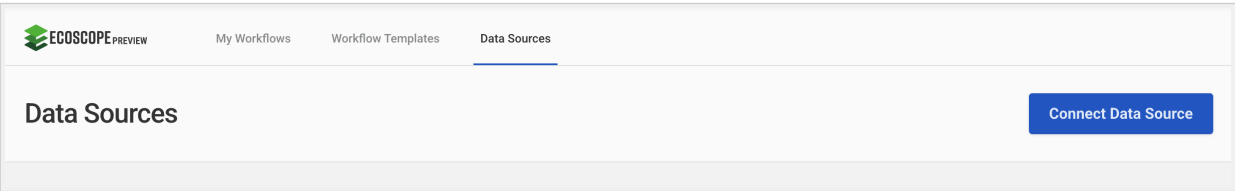
Prerequisites: Connect a Data Source

Before running the workflow you must have an **EarthRanger** data source configured in EcoScope Desktop. If you have already connected your EarthRanger site, skip ahead to **Step 1**. Otherwise follow the sub-steps below.

A

Open the Data Sources tab

Click **Data Sources** in the top navigation bar, then click **Connect Data Source**.



The Data Sources tab — click Connect Data Source to add a new connection.

B

Select EarthRanger

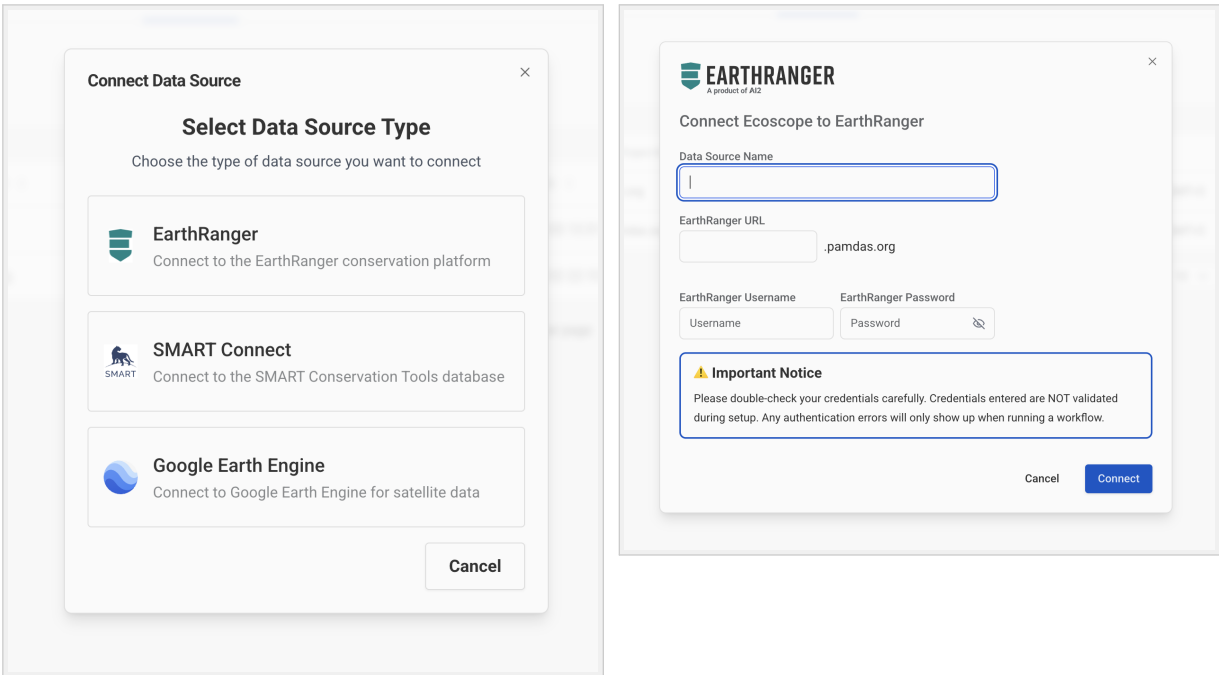
The **Select Data Source Type** dialog appears. Click **EarthRanger** to connect to the EarthRanger conservation platform.

C

Enter your EarthRanger credentials

Fill in the four fields and click **Connect**:

Field	Description
Data Source Name	A short label for this connection (e.g. TAAF)
EarthRanger URL	Your site subdomain — the part before .pamdas.org
EarthRanger Username	Your EarthRanger login username
EarthRanger Password	Your EarthRanger login password



Select EarthRanger as the data source type.

Enter your EarthRanger URL, username, and password.

Important: credentials are not validated during setup — any authentication errors will only appear when running a workflow.

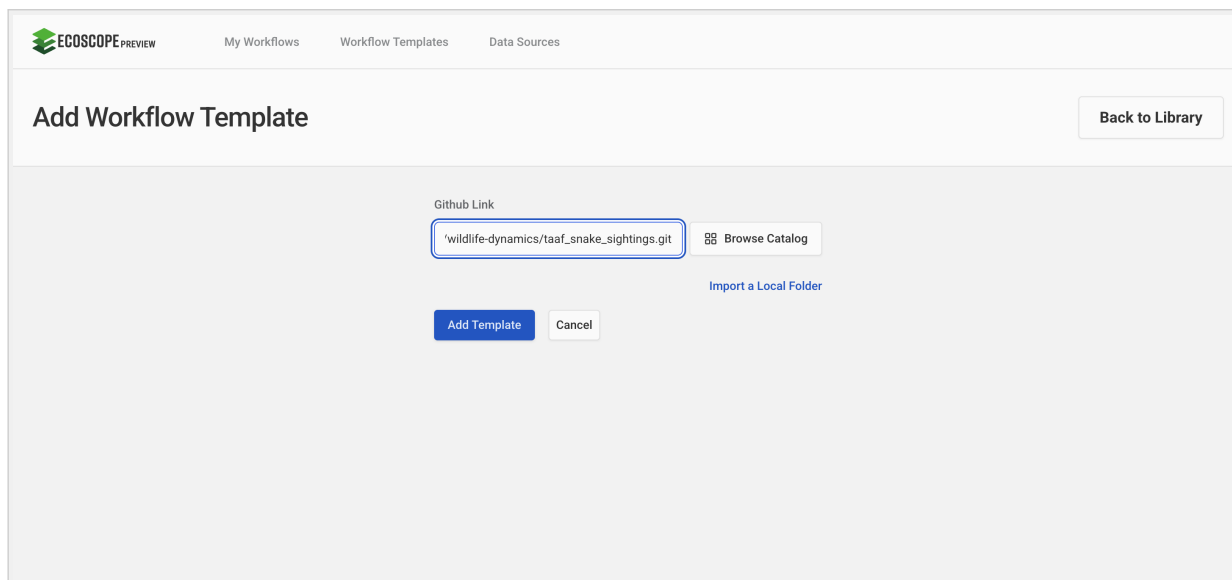
Step-by-Step Configuration

1 Add the Workflow Template

Click the **Workflow Templates** tab, then click **Add Workflow Template**. The **Add Workflow Template** page appears with a **Github Link** field. Paste the repository URL:

```
https://github.com/wildlife-dynamics/taaf_snake_sightings.git
```

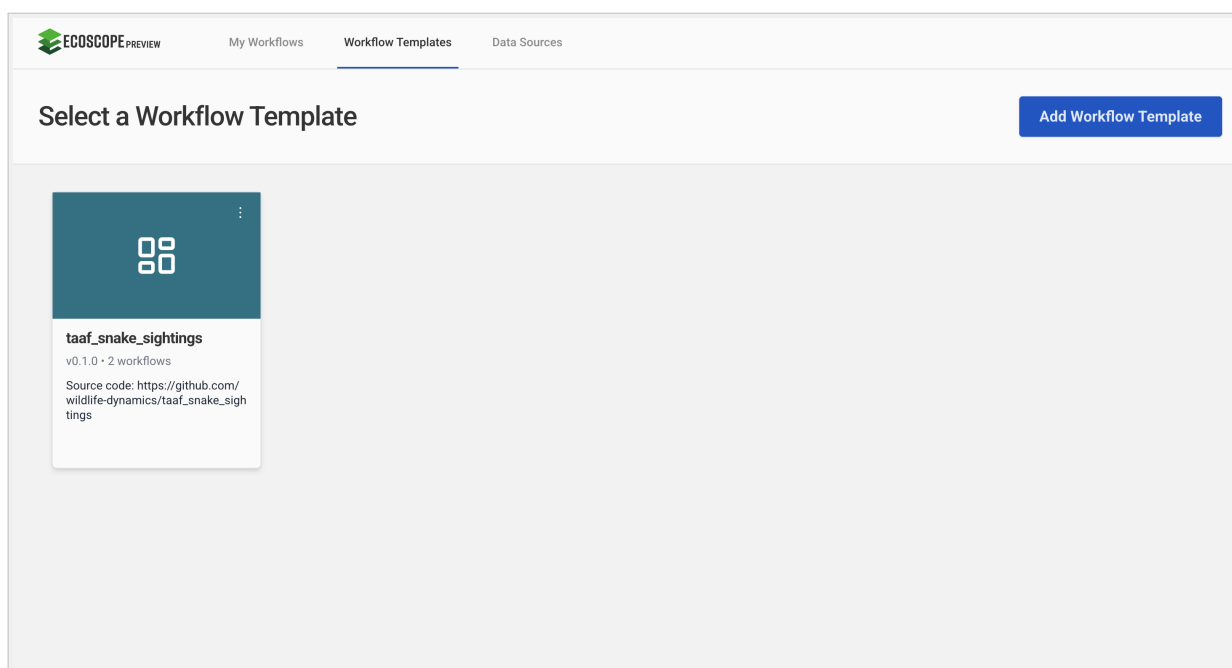
Then click **Add Template**.



Pasting the taaf_snake_sightings GitHub URL into the Add Workflow Template page.

2 Select the Workflow

Once the template has been added, the **taaf_snake_sightings** card appears in the Workflow Templates list. Click the card to open the configuration form.



The taaf_snake_sightings workflow card in the Workflow Templates list.

3 Set Workflow Details

The **Configure New Workflow** page opens with three sections. Fill in the **Workflow Name** (required) and optionally a **Workflow Description** to identify this run.

Set workflow details — Workflow Name and optional Description.

4 Set the Time Range, Connect to EarthRanger, and Submit

Scroll down to complete the remaining two sections:

Section	Field	Description
Define analysis time range	Timezone	Select local timezone (e.g. Africa/Nairobi UTC+03:00)
Define analysis time range	Since	Start date and time of the analysis period
Define analysis time range	Until	End date and time of the analysis period
Connect to EarthRanger	Data Source	Select your configured EarthRanger connection (e.g. TAAF)

Once all fields are filled, click **Submit**. The runner fetches observations, generates maps, builds the Word report, and publishes the dashboard. Per-species map generation scales with the number of species and may take several minutes.

Define time range, select the TAAF EarthRanger data source, then click Submit.

Workflow Steps Reference

Every task executed by the workflow in order.

Step	Purpose
Set Workflow Details	Records workflow name and description metadata.
Define Time Range	Sets the observation date window.
Connect to EarthRanger	Authenticates and opens a connection to the data source.
Fetch iNaturalist Observations	Pulls snake events of type <code>inat_observation</code> .
Parse Event Details	Extracts <code>taxon_name</code> , <code>common_name</code> , <code>user_name</code> , <code>inat_id</code> from raw JSON.
Filter to Valid Species	Keeps binomial names only; removes spatial outliers via IQR.
Build Species Summary Table	Computes per-species stats: count, observers, locations, date range, range km ² .
Save Summary Table as Word Doc	Exports summary table as formatted .docx.
Render Summary Table as HTML	Creates interactive sortable HTML table for the dashboard.
Prepare All-Species Points GDF	Builds combined GeoDataFrame with convex hull overlay.
Draw All-Species Map	Renders interactive HTML map with terrain and hillshade layers.
Create Observation Effort Grid	Aggregates observations into 15 km grid; colours by count.
Draw Observation Effort Map	Renders the effort choropleth as interactive HTML.
Create Species Richness Grid	Counts unique species per 15 km cell; colours by richness.
Draw Species Richness Map	Renders the richness choropleth as interactive HTML.
Generate All Per-Species Outputs	For every species: GeoDataFrame → HTML map → PNG → widget.
Build Per-Species Report Pages	Lays out per-species PNGs in a 2-col × 3-row grid per page.
Generate Word Report Template	Creates base Word document with logos, headings, and placeholders.
Render Main Report Sections	Screenshots the 3 HTML maps via Playwright; embeds in Word.
Assemble Final Report	Appends summary table and per-species pages → final .docx.
Snake Sightings Dashboard	Publishes all widgets — community maps and per-species maps.

Outputs

All outputs are written to the directory set by `ECOSCOPE_WORKFLOWS_RESULTS` (configured automatically by EcoScope Desktop).

Word Report

File	Description
taaf_snake_distribution_report.docx	Final combined report — cover, maps, summary table, per-species pages.
report_template.docx	Intermediate: base template with logos and placeholders.
snake_distribution_report_.docx	Intermediate: rendered cover + maps section.
species_summary_table.docx	Intermediate: summary table section.
per_species_maps.docx	Intermediate: per-species distribution pages.

Interactive Maps (HTML)

File	Description
*_all_snakes_distribution.html	Interactive all-species distribution map.
*_observation_effort.html	Interactive observation effort choropleth.
*_species_richness.html	Interactive species richness choropleth.
map_.html	One interactive map per species.

Dashboard Widgets

Widget	Description
Species Summary Table	Sortable table of all species statistics.
All Snake Species Distribution	Community-level distribution map widget.
Snake Observation Effort	15 km effort grid choropleth widget.
Species Richness Map	15 km richness grid choropleth widget.
Per-Species Maps	One widget per species, generated dynamically at runtime.

For support, open an issue on the [wildlife-dynamics/taaf_snake_sightings](#) GitHub repository.